Unit 21: Applied Optimization — Full Solutions

Warm-up

1. Two positive numbers, sum $= 20$, maximize the product.

Let the numbers be x and $20-x$. Objective: $P(x) = x(20-x) = 20x - x^2$.

$$P'(x) = 20 - 2x$$

Set $= 0$: $x = 10$. Check: $P''(x) = -2 < 0$, confirming a maximum.

Answer: the numbers are 10 and 10; the maximum product is 100.

2. Rectangle, perimeter $= 40$ m, maximize area.

Let the sides be x and y, with $2x + 2y = 40 \Rightarrow y = 20 - x$. Objective: $A(x) = x(20-x) = 20x - x^2$.

$$A'(x) = 20 - 2x$$

Set $= 0$: $x = 10$, so $y = 10$. $A''(x) = -2 < 0$ confirms a maximum.

Answer: the rectangle is a 10 m $\times$ 10 m square; maximum area $= 100$ m².

3. Two positive numbers, sum $= 24$, minimize the sum of their squares.

Let the numbers be x and $24-x$. Objective: $S(x) = x^2 + (24-x)^2$.

$$S'(x) = 2x + 2(24-x)(-1) = 2x - 2(24-x) = 4x - 48$$

Set $= 0$: $x = 12$, so $24 - x = 12$. $S''(x) = 4 > 0$ confirms a minimum.

Answer: the numbers are 12 and 12; the minimum sum of squares is $144 + 144 = 288$.

4. Farmer, 80 m fencing, rectangular pen (all 4 sides).

Let $2x + 2y = 80 \Rightarrow y = 40 - x$. Objective: $A(x) = x(40-x) = 40x - x^2$.

$$A'(x) = 40 - 2x$$

Set $= 0$: $x = 20$, so $y = 20$. $A''(x) = -2 < 0$ confirms a maximum.

Answer: a 20 m $\times$ 20 m square; maximum area $= 400$ m².

5. Rectangle, perimeter $= 60$, maximize area.

Let $2x + 2y = 60 \Rightarrow y = 30 - x$. Objective: $A(x) = x(30-x) = 30x - x^2$.

$$A'(x) = 30 - 2x$$

Set $= 0$: $x = 15$, so $y = 15$. $A''(x) = -2 < 0$ confirms a maximum.

Answer: a 15×15 square; maximum area $= 225$.

6. Two positive numbers, product $= 36$, minimize the sum.

Let the numbers be x and $\frac{36}{x}$. Objective: $S(x) = x + \frac{36}{x}$.

$$S'(x) = 1 - \frac{36}{x^2}$$

Set $= 0$: $x^2 = 36 \Rightarrow x = 6$ (taking the positive root). Then $\frac{36}{6} = 6$. $S''(x) = \frac{72}{x^3} > 0$ at $x = 6$, confirming a minimum.

Answer: the numbers are 6 and 6; the minimum sum is 12.

Standard

7. 200 m fencing, 3 sides (river on the 4th).

Let x be each of the two perpendicular sides, and y be the side parallel to the river. Constraint: $2x + y = 200 \Rightarrow y = 200 - 2x$. Objective: $A(x) = xy = x(200 - 2x) = 200x - 2x^2$.

$$A'(x) = 200 - 4x$$

Set $= 0$: $x = 50$, so $y = 200 - 100 = 100$. Domain: $0 < x < 100$. $A''(x) = -4 < 0$ confirms a maximum.

Answer: the sides perpendicular to the river should be 50 m, and the side parallel to the river should be 100 m; maximum area $= 5000$ m^2.

8. Square cardboard, side 12 in, cut x from corners.

The resulting open box has base side $(12 - 2x)$ and height x. Objective: $V(x) = x(12 - 2x)^2$, for $0 < x < 6$.

Expand: $V(x) = x(144 - 48x + 4x^2) = 144x - 48x^2 + 4x^3$.

$$V'(x) = 144 - 96x + 12x^2 = 12(x^2 - 8x + 12) = 12(x - 2)(x - 6)$$

Critical points: $x = 2$, $x = 6$. Only $x = 2$ lies inside the open domain $(0, 6)$.

Testing: for $x < 2$ (e.g., $x = 1$): $12(-1)(-5) = 60 > 0$ (increasing). For $2 < x < 6$ (e.g., $x = 3$): $12(1)(-3) = -36 < 0$ (decreasing). Sign changes $+$ to $-$, confirming a maximum at $x = 2$.

$$V(2) = 2(12 - 4)^2 = 2(8)^2 = 2(64) = 128$$

Answer: cut squares of side $x = 2$ inches; the resulting box has base 8×8 inches and height 2 inches, giving a maximum volume of 128 in^3.

9. Rectangular garden, area $= 200$ ft^2, minimize the perimeter.

Let $xy = 200 \Rightarrow y = \frac{200}{x}$. Objective: $P(x) = 2x + 2y = 2x + \frac{400}{x}$.

$$P'(x) = 2 - \frac{400}{x^2}$$

Set $= 0$: $x^2 = 200 \Rightarrow x = \sqrt{200} = 10\sqrt{2}$. Then $y = \dfrac{200}{10\sqrt{2}} = \dfrac{20}{\sqrt{2}} = 10\sqrt{2}$.

Answer: the garden should be a $10\sqrt{2} \times 10\sqrt{2}$ ft square (both dimensions equal, ≈ 14.14 ft); minimum perimeter $= 40\sqrt{2} \approx 56.57$ ft.

10. Minimize $f(x) = x + \dfrac{1}{x}$ for $x > 0$.

$$f'(x) = 1 - \frac{1}{x^2}$$

Set $= 0$: $x^2 = 1 \Rightarrow x = 1$ (positive root). $f''(x) = \dfrac{2}{x^3} > 0$ at $x = 1$, confirming a minimum.

$$f(1) = 1 + 1 = 2$$

Answer: the minimum value is 2, occurring at $x = 1$.

11. Box, square base, no top, volume $= 32$ ft^3, minimize surface area.

Let the base side be x and height h. Constraint: $x^2 h = 32 \Rightarrow h = \dfrac{32}{x^2}$.

Objective (base + 4 sides, no top): $S(x) = x^2 + 4xh = x^2 + 4x\left(\dfrac{32}{x^2}\right) = x^2 + \dfrac{128}{x}$.

$$S'(x) = 2x - \frac{128}{x^2}$$

Set $= 0$: $2x^3 = 128 \Rightarrow x^3 = 64 \Rightarrow x = 4$. Then $h = \dfrac{32}{16} = 2$.

$S''(x) = 2 + \dfrac{256}{x^3} > 0$ for all $x > 0$, confirming a minimum.

$$S(4) = 16 + \frac{128}{4} = 16 + 32 = 48$$

Answer: base 4×4 ft, height 2 ft; minimum surface area $= 48$ ft^2.

12. Two positive numbers, product $= 100$, minimize (first number) $+2\times$(second number).

Let $xy = 100 \Rightarrow y = \dfrac{100}{x}$. Objective: $S(x) = x + 2y = x + \dfrac{200}{x}$.

$$S'(x) = 1 - \frac{200}{x^2}$$

Set $= 0$: $x^2 = 200 \Rightarrow x = \sqrt{200} = 10\sqrt{2}$. Then $y = \dfrac{100}{10\sqrt{2}} = \dfrac{10}{\sqrt{2}} = 5\sqrt{2}$.

$S''(x) = \dfrac{400}{x^3} > 0$, confirming a minimum.

Answer: the numbers are $x = 10\sqrt{2}$ and $y = 5\sqrt{2}$; the minimum value of $x + 2y$ is $10\sqrt{2} + 2(5\sqrt{2}) = 20\sqrt{2} \approx 28.28$.

13. Highway on one side, 300 ft fencing for the other 3 sides.

Let x be each of the two perpendicular sides, y the side parallel to the highway. $2x + y = 300 \Rightarrow y = 300 - 2x$. Objective: $A(x) = x(300 - 2x) = 300x - 2x^2$.

$$A'(x) = 300 - 4x$$

Set $= 0$: $x = 75$, so $y = 300 - 150 = 150$. $A''(x) = -4 < 0$ confirms a maximum.

Answer: the perpendicular sides should be 75 ft each, and the side parallel to the highway should be 150 ft; maximum area = 11,250 ft².

Challenge

14. Farmland, 400 m wire, river on one side.

Let x be each of the two perpendicular sides, y the side parallel to the river. $2x + y = 400 \Rightarrow y = 400 - 2x$. Objective: $A(x) = x(400 - 2x) = 400x - 2x^2$.

$$A'(x) = 400 - 4x$$

Set $= 0$: $x = 100$, so $y = 400 - 200 = 200$. Domain: $0 < x < 200$. $A''(x) = -4 < 0$ confirms a maximum.

Answer: the sides perpendicular to the river should be 100 m each, and the side parallel to the river should be 200 m; maximum enclosed area = 20,000 m².

15. Closed cylinder, volume = 500 cm³, minimize surface area.

Constraint: $V = \pi r^2 h = 500 \Rightarrow h = \dfrac{500}{\pi r^2}$.

Objective (top + bottom + side): $S(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \left(\dfrac{500}{\pi r^2}\right) = 2\pi r^2 + \dfrac{1000}{r}$.

$$S'(r) = 4\pi r - \frac{1000}{r^2}$$

Set $= 0$: $4\pi r^3 = 1000 \Rightarrow r^3 = \dfrac{250}{\pi} \Rightarrow r = \sqrt[3]{\dfrac{250}{\pi}} \approx 4.30$ cm.

Then $h = \dfrac{500}{\pi r^2}$. Using the relationship $\pi = \dfrac{250}{r^3}$ from above:

$$h = \frac{500}{\left(\frac{250}{r^3}\right) r^2} = \frac{500}{\frac{250}{r}} = \frac{500r}{250} = 2r$$

Answer: $r = \sqrt[3]{\dfrac{250}{\pi}} \approx 4.30$ cm, and $h = 2r \approx 8.60$ cm — notice the elegant result that the minimizing height is always exactly twice the radius for a closed cylinder.

16. Point on $y = 2x + 3$ closest to the origin.

Distance squared from a point $(x, 2x+3)$ on the line to the origin:

$$D(x) = x^2 + (2x+3)^2 = x^2 + 4x^2 + 12x + 9 = 5x^2 + 12x + 9$$

(Minimizing $D(x)$ is equivalent to minimizing the actual distance, since square-rooting doesn't change where the minimum occurs.)

$$D'(x) = 10x + 12$$

Set $= 0$: $x = -\dfrac{12}{10} = -\dfrac{6}{5}$. Then $y = 2\left(-\dfrac{6}{5}\right) + 3 = -\dfrac{12}{5} + \dfrac{15}{5} = \dfrac{3}{5}$.

$D''(x) = 10 > 0$ confirms a minimum.

Answer: the closest point is $\left(-\dfrac{6}{5}, \dfrac{3}{5}\right)$. (The minimum distance itself is $\sqrt{D\left(-\frac{6}{5}\right)} = \sqrt{1.8} = \dfrac{3\sqrt{5}}{5} \approx 1.34$.)

17. Box, square base, no top, volume = 108 in^3, minimize material.

Let base side x, height h. Constraint: $x^2 h = 108 \Rightarrow h = \dfrac{108}{x^2}$.

Objective: $S(x) = x^2 + 4xh = x^2 + 4x\left(\dfrac{108}{x^2}\right) = x^2 + \dfrac{432}{x}$.

$$S'(x) = 2x - \frac{432}{x^2}$$

Set $= 0$: $2x^3 = 432 \Rightarrow x^3 = 216 \Rightarrow x = 6$. Then $h = \dfrac{108}{36} = 3$.

$S''(x) = 2 + \dfrac{864}{x^3} > 0$, confirming a minimum.

$$S(6) = 36 + \frac{432}{6} = 36 + 72 = 108$$

Answer: base 6×6 in, height 3 in; minimum material used = 108 in^2.

18. Norman window, perimeter = 20 ft, maximize area.

Let x be the rectangle's width (also the semicircle's diameter), and y the rectangle's height.

Perimeter constraint: bottom (x) + two sides $(2y)$ + semicircular arc ($\dfrac{\pi x}{2}$, since the arc length of a full circle of radius $\frac{x}{2}$ is πx, and we only use half):

$$x + 2y + \frac{\pi x}{2} = 20$$

Solve for y:

$$2y = 20 - x - \frac{\pi x}{2} \quad \Rightarrow \quad y = 10 - \frac{x}{2} - \frac{\pi x}{4}$$

Area: rectangle plus semicircle ($\frac{1}{2}\pi\left(\frac{x}{2}\right)^2 = \frac{\pi x^2}{8}$):

$$A(x) = xy + \frac{\pi x^2}{8}$$

Substitute y:

$$A(x) = x\left(10 - \frac{x}{2} - \frac{\pi x}{4}\right) + \frac{\pi x^2}{8} = 10x - \frac{x^2}{2} - \frac{\pi x^2}{4} + \frac{\pi x^2}{8}$$

Combine the πx^2 terms: $-\frac{\pi x^2}{4} + \frac{\pi x^2}{8} = -\frac{\pi x^2}{8}$.

$$A(x) = 10x - \frac{x^2}{2} - \frac{\pi x^2}{8} = 10x - \frac{(4+\pi)x^2}{8}$$

Differentiate:

$$A'(x) = 10 - \frac{(4+\pi)x}{4}$$

Set $= 0$:

$$10 = \frac{(4+\pi)x}{4} \quad \Rightarrow \quad x = \frac{40}{4+\pi} \approx 5.60 \text{ ft}$$

Then:

$$y = 10 - \frac{x}{2} - \frac{\pi x}{4}$$

It turns out (after simplifying algebraically) that at this optimal x, $y = \frac{x}{2}$ exactly:

$$y = \frac{20}{4+\pi} \approx 2.80 \text{ ft}$$

The maximum area simplifies to:

$$A = \frac{200}{4+\pi} \approx 28.01 \text{ ft}^2$$

Answer: width $x = \frac{40}{4+\pi} \approx 5.60$ ft, rectangle height $y = \frac{20}{4+\pi} \approx 2.80$ ft; maximum area ≈ 28.01 ft^2.

Applied

19. $q = 1200 - 40p$, maximize $R = pq$.

$$R(p) = p(1200 - 40p) = 1200p - 40p^2$$

$$R'(p) = 1200 - 80p$$

Set $= 0$: $p = 15$. $R''(p) = -80 < 0$ confirms a maximum. $q = 1200 - 40(15) = 1200 - 600 = 600$.

Answer: set the price at $p = \$15$; maximum revenue $= 15 \times 600 = \$9000$.

20. $C(x) = 2x + \dfrac{800}{x}$, minimize for $x > 0$.

$$C'(x) = 2 - \frac{800}{x^2}$$

Set $= 0$: $x^2 = 400 \Rightarrow x = 20$. $C''(x) = \dfrac{1600}{x^3} > 0$ at $x = 20$, confirming a minimum.

$$C(20) = 40 + 40 = 80$$

Answer: order size $x = 20$ units; minimum total cost $= \$80$.

21. Storage yard, wall on one side, 240 ft fencing for the other 3 sides.

Let x be each of the two perpendicular sides, y the side parallel to the wall. $2x + y = 240 \Rightarrow y = 240 - 2x$. Objective: $A(x) = x(240 - 2x) = 240x - 2x^2$.

$$A'(x) = 240 - 4x$$

Set $= 0$: $x = 60$, so $y = 240 - 120 = 120$. $A''(x) = -4 < 0$ confirms a maximum.

Answer: the sides perpendicular to the wall should be 60 ft each, and the side parallel to the wall should be 120 ft; maximum area = 7200 ft^2.

22. $P(x) = -0.02x^2 + 40x - 300$.

$$P'(x) = -0.04x + 40$$

Set $= 0$: $x = 1000$. $P''(x) = -0.04 < 0$ confirms a maximum.

$$P(1000) = -0.02(1{,}000{,}000) + 40{,}000 - 300 = -20{,}000 + 40{,}000 - 300 = 19{,}700$$

Answer: produce 1000 units; maximum profit $= \$19{,}700$.